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1. Introduction — ACAS vs TCAS

ACAS = Airborne Collision Avoidance System (ICAO name)
TCAS = Traffic alert and Collision Avoidance System (US manufacturer name, now widely accepted alternative)
Four systems exist: TCAS I, II, III, IV (increasing levels of protection)

Regulatory Requirement: All commercial air transport turbine-powered aircraft registered in Europe with MTOM > **5700 kg** or > **19 passenger seats** must have at least ACAS II (version 7).

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2. TCAS I

First generation collision avoidance system. Capabilities:

- Detects and displays range and approximate relative bearing of nearby aircraft
- If intruder carries Mode C: relative altitude also displayed
- Generates **Traffic Advisories (TAs) only**: "Traffic, Traffic"
- Does **NOT** give Resolution Advisories (no recommended manoeuvre)

3. TCAS II — Principle

TCAS II detects intruders, assesses collision risk, and presents both TAs and RAs. Uses Mode S data link to co-ordinate with other TCAS-equipped aircraft, providing **complementary vertical avoidance** instructions.

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Operates on secondary radar principle using SSR frequencies of **1030 MHz** (interrogation) and **1090 MHz** (reply) in an **air-to-air role**.

Creates two protective 3D bubbles around the aircraft:

- **Outer bubble (caution area)** → Traffic Advisory (TA) zone
- **Inner bubble (warning area)** → Resolution Advisory (RA) zone

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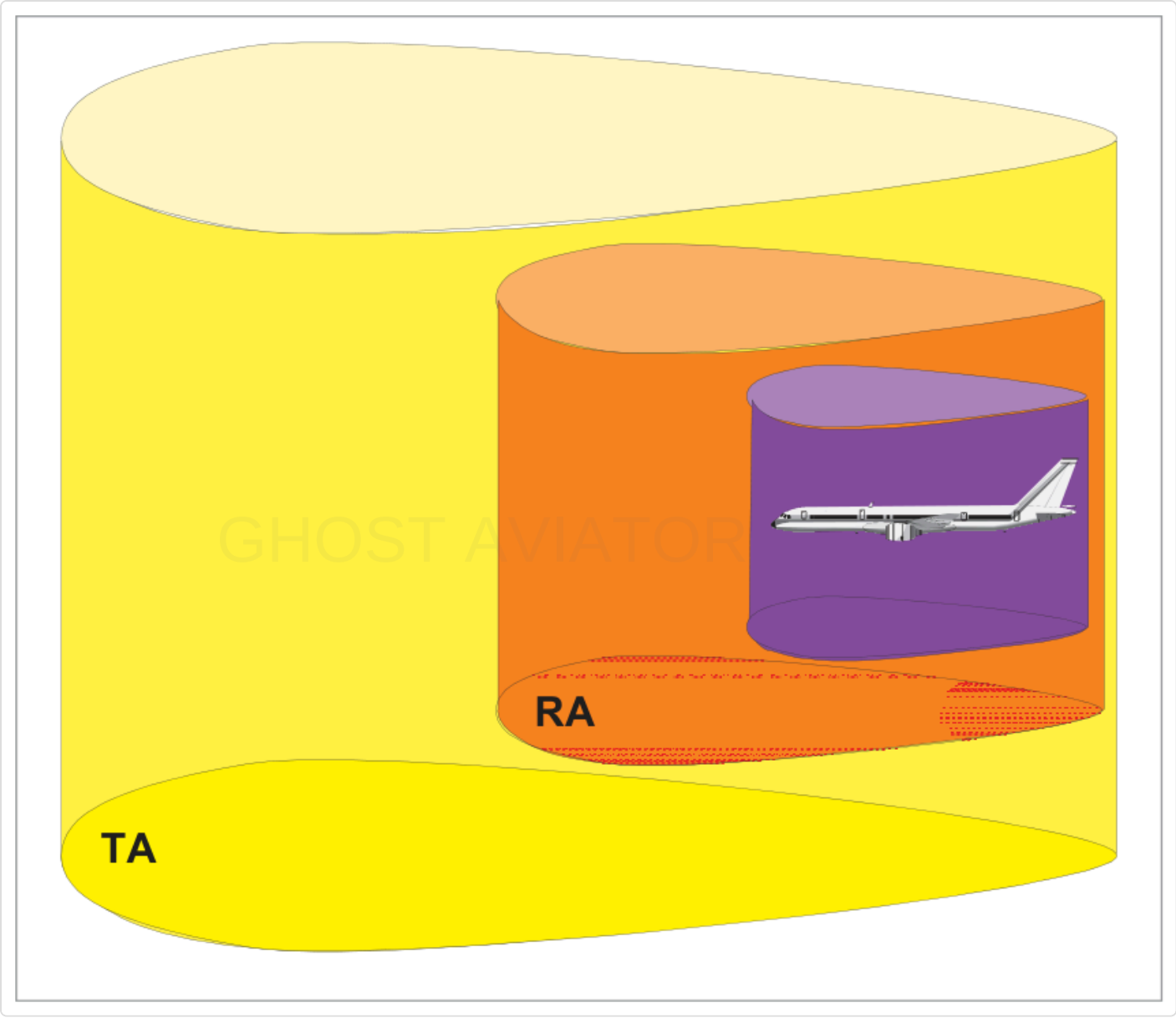


Fig 35.1 — TCAS II two protective 3D bubbles

Key TCAS II Limitation: TCAS II never offers collision avoidance commands in the horizontal plane — **vertical plane only** (climb or descend).

4. Aircraft Equipment (Transponder Requirements)

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Transponder Type	Information Available	TCAS Capability
Mode A	Identity only (no height)	2D — TAs only
Mode C	Identity + height	3D — TAs and RAs
Mode S	Identity + height + discrete data link	3D + co-ordinated RAs between TCAS aircraft
None / off	—	Invisible to TCAS — collision risk!

5. Operation — Range, Bearing, Height

Range

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Determined by measuring time between interrogation transmission and response (radar principle). Interrogation approximately **once per second**. In densely populated airspace: **reduced surveillance** — interrogation every **5 seconds**, priority to closest aircraft.

- Normal detection range: **~30 NM**
- Reduced surveillance range: **~5 NM**

Bearing

Determined by directional antenna. Due to wavelengths and antenna size, **bearing is the least accurate parameter**.

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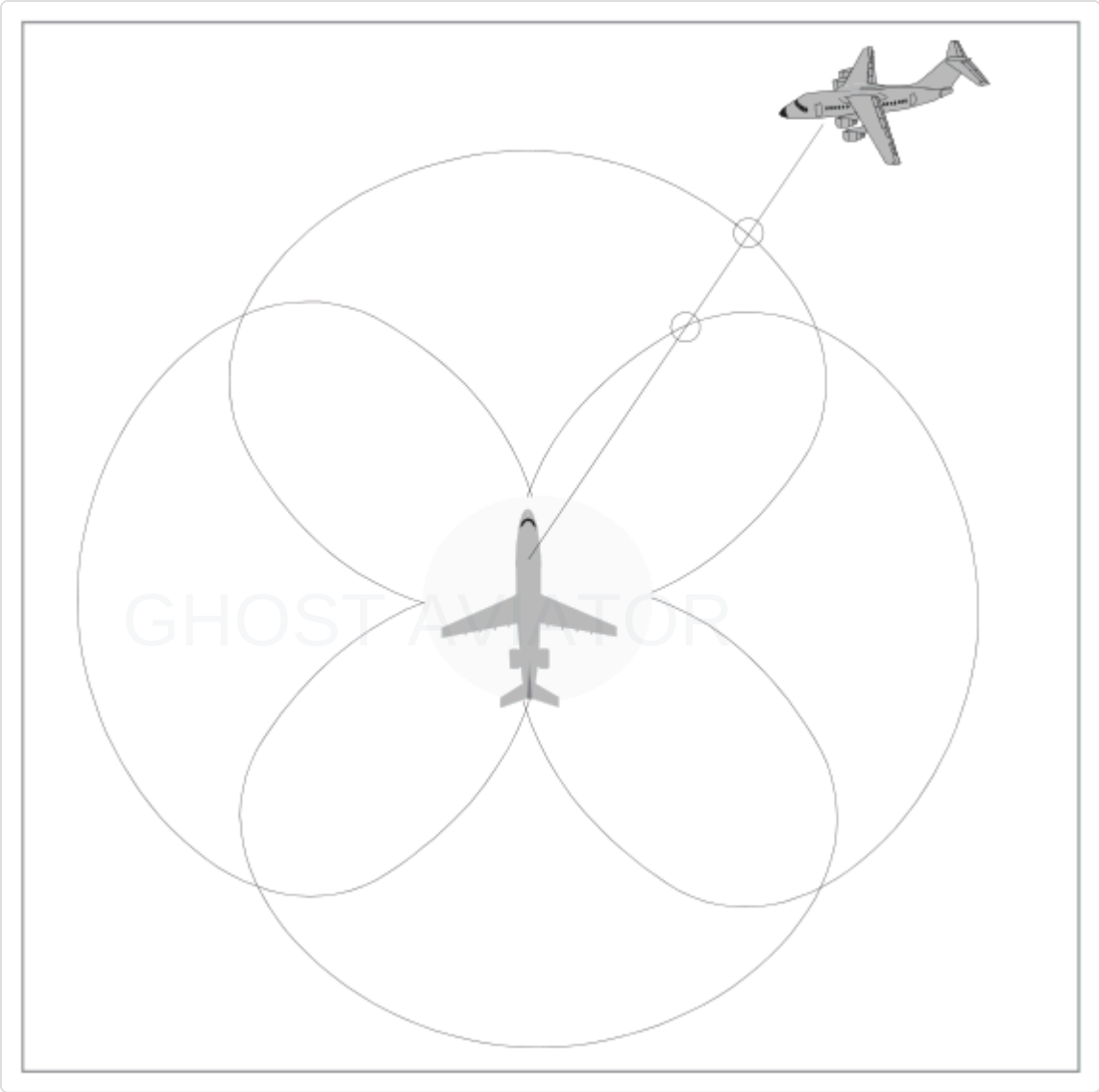


Fig 35.2 — TCAS bearing determination using directional antenna

Height

Found by comparing intruder's Mode C altitude with TCAS aircraft's own altitude.

6. System Interconnections

TCAS feeds from (and limitations based on):

- **Radio altimeter:** Modifies RAs near the ground:
  - No instructions at all below **400 ft AGL**
  - No descent RAs below **1000 ft AGL**
  - No increase rate of descent below **1450 ft AGL**
- **Aircraft configuration (gear/flap):** Poor climb performance → TCAS avoids climb demands
- **ADC:** At high altitude near performance ceiling → TCAS avoids climbing RA

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- **IRS:** Inertial vertical acceleration input

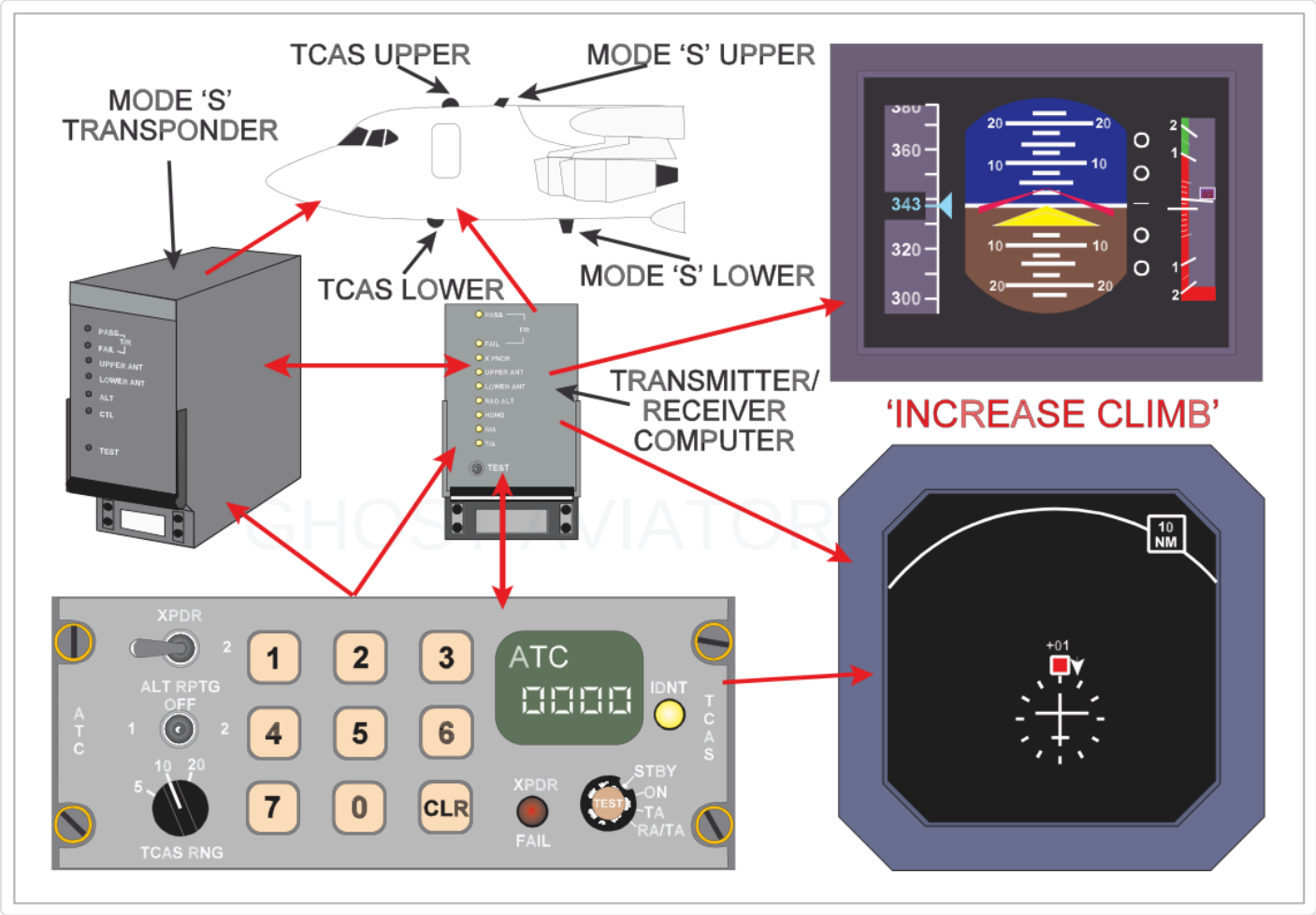


Fig 35.3 — TCAS system installation in a commuter/feeder airliner

7. Synthetic Voice Prioritization

Synthetic voice priority order (highest first):

1. Stall Identification/Prevention (stick shake/push) — voice inhibited during stick push
2. Windshear — inhibits both GPWS and TCAS
3. GPWS — inhibits TCAS
4. TCAS (lowest priority)

8. Traffic Advisories (TAs) and Resolution Advisories (RAs)

TCAS can monitor up to **30 aircraft** simultaneously on display. Conflict assessment based on tau (τ) = Closest Point of Approach (CPA).

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Advisory Type	Bubble Penetrated	Time to Collision	Symbol	Voice
TA (Traffic Advisory)	Outer (caution) bubble	**45–35 seconds**	Solid amber circle	"Traffic, Traffic"
RA (Resolution Advisory)	Inner (warning) bubble	**35–15 seconds**	Solid red rectangle	Climb/Descend commands

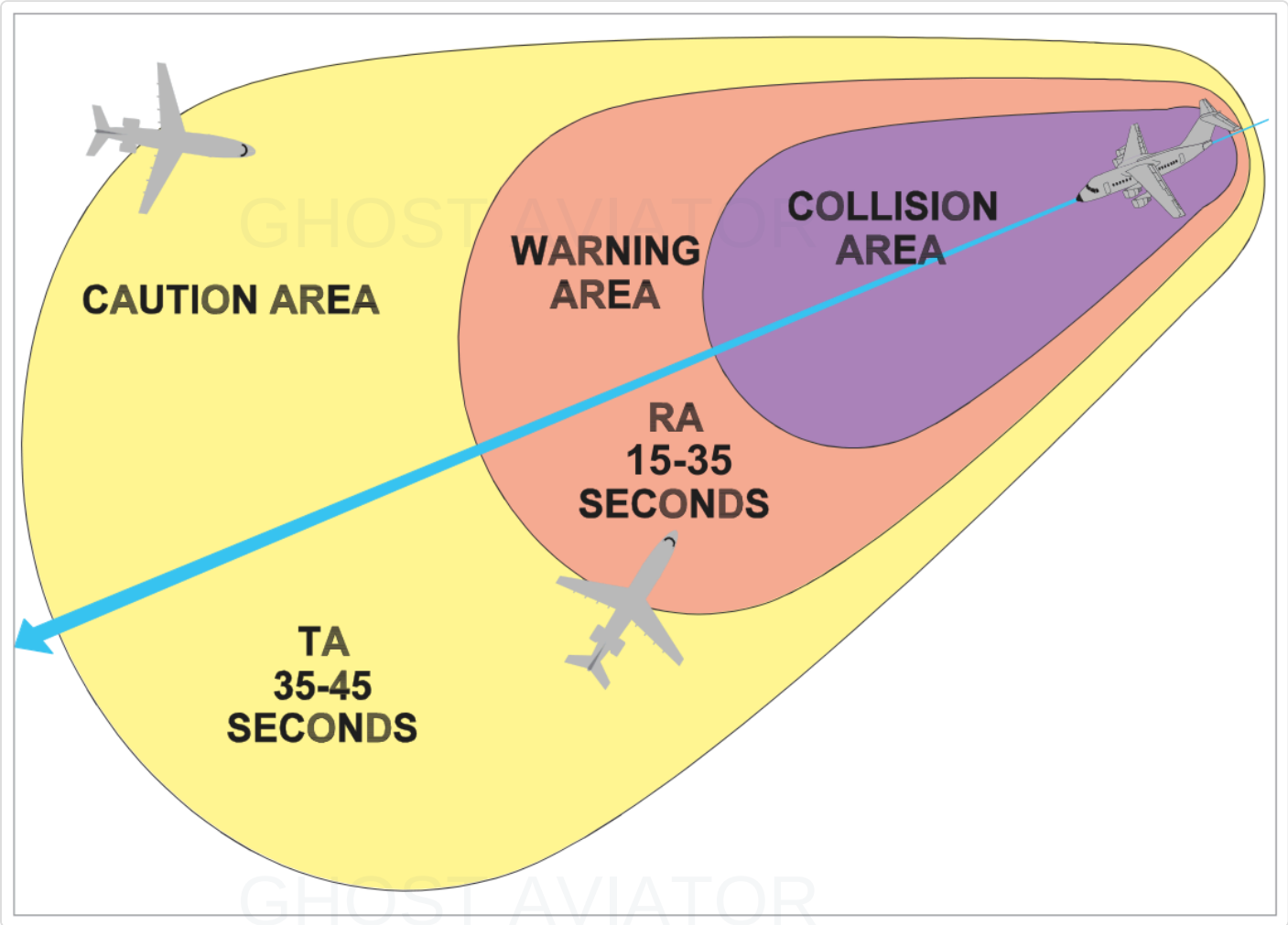


Fig 35.4 — TCAS TA and RA time-to-collision zones

9. Resolution Advisory Types

RA Type	Situation	Voice
Preventative RA	No collision risk unless one aircraft changes level	"Monitor Vertical Speed"
Corrective RA	Collision risk exists; manoeuvre required	"Climb" "Descend" "Increase Climb" "Decrease Descent" etc.

Separation intended with smooth RA response: 300–600 ft vertical (700 ft above FL410). Separation varies with Sensitivity Level (SL) which decreases with altitude.

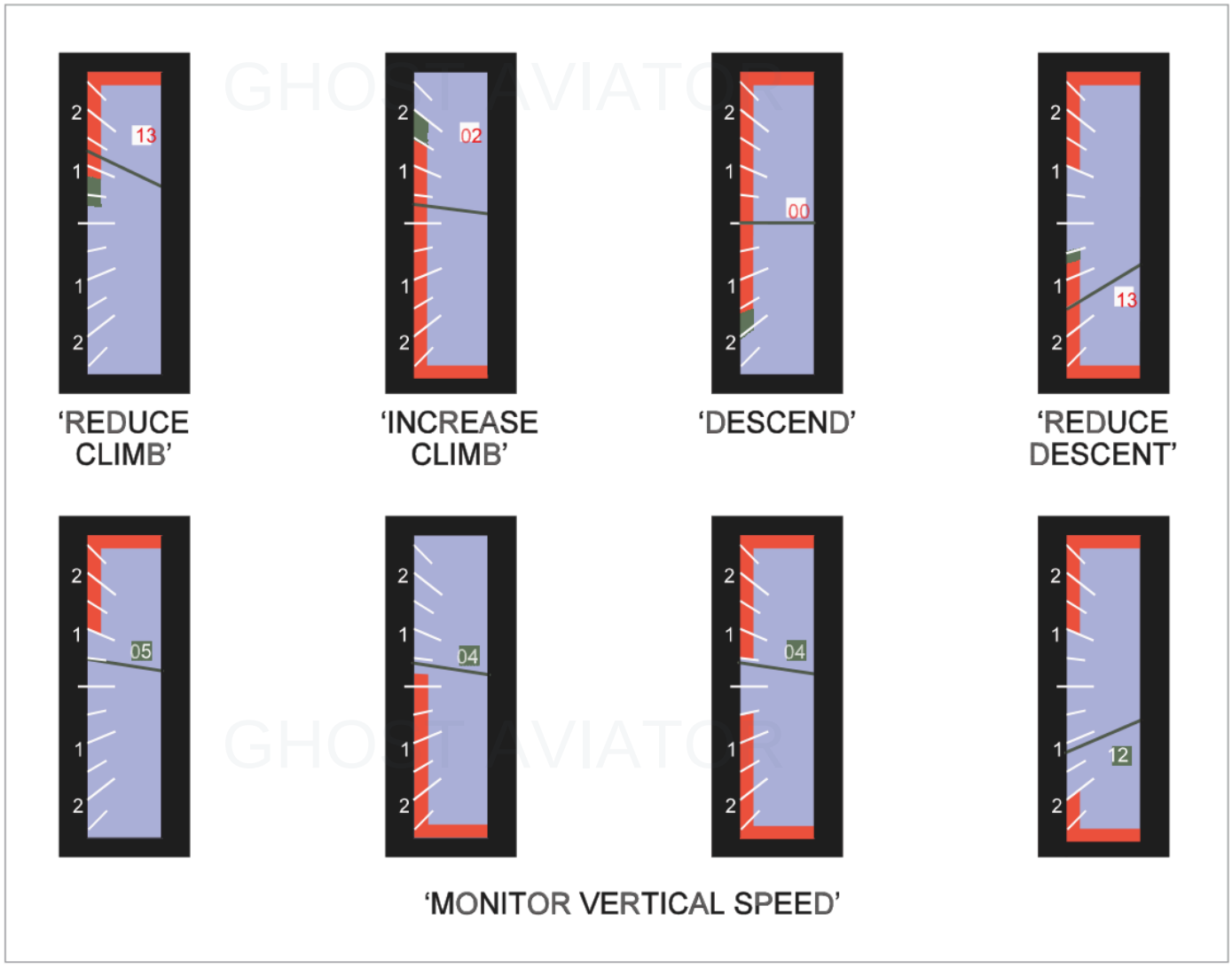


Fig 35.5 — Preventative and corrective RAs displayed on PFD vertical speed tape

10. Traffic Symbols

Symbol	Appearance	Meaning	Range / Height
RA Traffic	Solid red rectangle	Serious collision threat — manoeuvre immediately	Inner bubble
TA Traffic	Solid amber circle	Potential collision threat — prepare for RA	Outer bubble
Proximate Traffic	Solid cyan diamond	Near traffic, not a threat	Within **6 NM** and ±**1200 ft**
Other Traffic	Hollow cyan diamond	Transponder-equipped, within display range	Within range, ±**2700 ft** (or ±8700/2700 in climb/descent)

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11. Data Tag and Off-Scale Advisories

Data Tag (Mode C equipped intruders)

- Relative altitude in hundreds of feet
- + sign = intruder above; − sign = intruder below
- ↑ or ↓ trend arrow when vertical rate ≥ **500 fpm**
- No number/arrow = non-altitude-reporting intruder

Off-Scale Traffic Advisory

When intruder is outside display range but inside caution/warning area: half of the appropriate symbol appears at the correct bearing at the edge of the display with data tag (if room).

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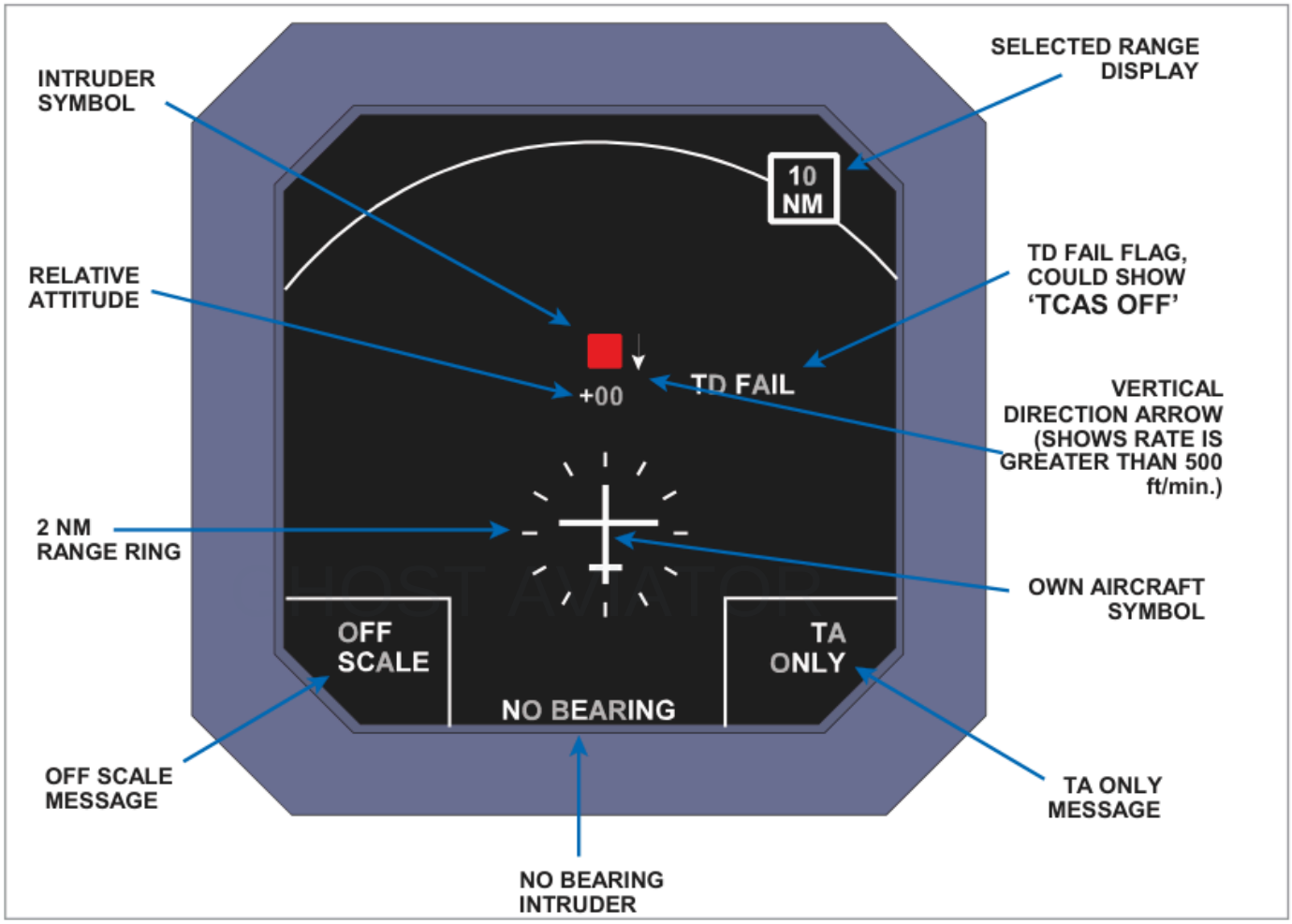


Fig 35.7 — TCAS on Electronic Vertical Speed Indicator

12. TCAS Displays and Control Panel

Function Switch Positions

Position	Function
STANDBY	Warm-up power only; system not operational
ON	Transponder only operational (visible to ACAS, but no TCAS function)
TA	Transponder + TCAS operational; TAs only generated. "TA ONLY" shown on display. Used for engine failure, reduced performance, parallel runway ops.
RA/TA	Full operation — both RAs and TAs generated (normal operation)
TEST	Full BITE check. Success: "TCAS SYSTEM TEST OK"; Failure: "TCAS SYSTEM TEST FAIL"

TCAS RNG Switch

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Selects display range: **5, 10, or 20 NM**. Does NOT alter detection range or warning thresholds.

13. No Bearing Advisories

If TCAS cannot track intruder bearing (e.g., antenna screening), RA or TA appears lower-centre of display in appropriate colour. Up to two lines of information. Example: "TA 2.2- 04 ↑" = TA 2.2 NM away, 400 ft below, climbing ≥500 fpm.

> No bearing information does NOT degrade TCAS ability to compute a TA or RA — only the display position is affected.

14. Action on Receiving TAs and RAs

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Advisory	Crew Action
TA	Assimilate information. Begin visual search. Prepare for RA. Seek ATC advice if traffic not seen. **Do NOT manoeuvre on TA alone.**
RA	**Initiate required manoeuvre immediately.** Adjust flight path, power, trim. Inform ATC of deviation as soon as possible.

Rules for disregarding RAs:

- Never manoeuvre OPPOSITE to RA (sense may have been co-ordinated with intruder via Mode S)
- RAs may be disregarded ONLY if pilot visually identifies the conflicting traffic AND decides no deviation is necessary
- If ATC instruction conflicts with RA — **FOLLOW THE RA**

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Quick Revision Summary — Chapter 35:

- ACAS = ICAO name; TCAS = manufacturer/common name. Mandatory >5700 kg or >19 seats (Europe)
- TCAS I: TAs only. TCAS II: TAs + RAs (vertical only, never horizontal)
- TCAS II frequencies: 1030 MHz (interrogate) / 1090 MHz (reply)
- Mode A: TAs only (2D). Mode C: TAs + RAs (3D). Mode S: TAs + RAs + co-ordinated RAs
- Interrogation: ~1/sec normal; every 5 sec reduced surveillance. Range: 30 NM normal / 5 NM reduced
- Bearing least accurate parameter (small antenna, SSR wavelength)
- RA height restrictions: no instructions <400 ft AGL; no descent RA <1000 ft; no increase descent <1450 ft
- TA = outer bubble = 45–35 sec = solid amber circle. RA = inner bubble = 35–15 sec = solid red rectangle
- Proximate traffic: solid cyan diamond; within 6 NM ±1200 ft
- Data tag ↑↓ when V/S ≥500 fpm; + = above; − = below
- RA types: Preventative ("Monitor VS") / Corrective ("Climb/Descend")
- Intended separation (smooth response): 300–600 ft; 700 ft above FL410
- On TA: look; prepare; do NOT manoeuvre. On RA: manoeuvre IMMEDIATELY; follow RA over ATC if conflict
- Voice priority: Stall → Windshear → GPWS → TCAS

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Practice Questions & Additional Q&A

Practice Questions (Chapter 35)

```
Answers: 1-a 2-b 3-a 4-c 5-d
```

Q1. On receipt of a TCAS RA your action is to:

a. Initiate the required manoeuvre immediately

b. Make a note of the details

c. Request a flight clearance deviation from ATC

d. Do nothing until a TA is received

Correct Answer: **(a)** Initiate the required manoeuvre immediately

Explanation: An RA indicates a serious collision threat (35–15 seconds to CPA). The only correct response is immediate execution of the required manoeuvre. See Section 14.

Q2. Which statement concerning TCAS is correct?

a. TCAS II provides avoidance instructions in vertical and horizontal planes

b. TCAS II cannot provide information on non-SSR equipped intruders

c. TCAS II requires Mode S to be fitted to other aircraft

d. TCAS II provides advice on which way to turn

Correct Answer: **(b)**

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Explanation: TCAS II relies on intruder SSR transponders. Aircraft without transponders are invisible. TCAS II only provides vertical avoidance (never horizontal). Mode S is needed for co-ordinated RAs but not required for basic detection (Mode A/C suffice). See Section 4.

Q3. The difference between TCAS I and TCAS II is:

a. TCAS II can provide TAs and RAs whilst TCAS I can only provide TAs

b. TCAS II is only for large aircraft >30 passengers

c. TCAS I requires Mode A only; TCAS II requires Mode C or S

d. TCAS II only works with EFIS equipped aircraft

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Correct Answer: **(a)**

Explanation: TCAS I = TAs only (alerts crew to look for traffic). TCAS II = TAs + RAs (provides vertical manoeuvre advice). See Sections 2 and 3.

Q4. The aural messages provided by TCAS II are:

a. Threat, Climb; Threat, Descend

b. Climb left; Climb right; Descend left; Descend right

c. Climb; Descend; Increase Climb; Increase Descent

d. Turn Left, Turn Right, Increase Turn, Decrease Turn

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Correct Answer: **(c)**

Explanation: TCAS II RA voice commands: "Climb" "Increase Climb" "Descend" "Increase Descent" "Monitor Vertical Speed" "Decrease Climb" "Decrease Descent" — all in the vertical plane only. No

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lateral (turn) instructions. See Section 3.

Q5. RAs may be disregarded only when:

a. RAs may be disregarded only when the pilot visually identifies the traffic, decides no deviation is necessary, and has clearance confirmed by ATC

b. RAs may be disregarded when they conflict with an ATC instruction

c. RAs must never be disregarded

d. RAs may be disregarded only when the pilot visually identifies the traffic and decides no deviation is necessary

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Correct Answer: **(d)**

Explanation: RAs may be disregarded ONLY when the pilot visually identifies the conflicting traffic AND decides no deviation is necessary. ATC confirmation is NOT required — in fact, if ATC instruction conflicts with RA, the RA takes precedence. See Section 14.

Instructor's Note: *This is one of the most exam-important rules. (a) is wrong because ATC confirmation is not required. (b) is wrong because ATC instruction that conflicts = follow the RA. (c) is wrong because there is one exception (visual identification).*

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Q6. TCAS will NOT give any avoidance commands when the aircraft is below:

a. 1000 ft AGL

b. 1450 ft AGL

c. 400 ft AGL

d. 200 ft AGL

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Correct Answer: **(c)** 400 ft AGL

Explanation: TCAS gives no instructions at all below **400 ft AGL**. Below 1000 ft: no descent RAs. Below 1450 ft: no increase rate of descent commands. The most restrictive is 400 ft — no instructions at all. See Section 6.

Instructor's Note: *Three altitude limits to memorise: 400 ft (nothing), 1000 ft (no descent), 1450 ft (no increase descent). On approach these prevent TCAS from commanding dangerous go-arounds below decision height.*

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